

EDUCATION

National University of Singapore

College of Humanities and Sciences · (Aug 2022 - May 2026)

- Bachelor of Science: (Hons) Double Major in Mathematics (1st) and Statistics (2nd), with Minor in Computer Science
 - GPA: 4.49/5.00
 - Awards: AY25/26 Semester 1 Dean's List
 - Relevant Coursework: Optimization for Large-scale Data-driven Inference, Data Modelling and Computation, Applied Time Series Analysis, Nonlinear Programming, Statistical Computing, Mathematical Statistics, Software Engineering
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PROJECTS

Simulation-based Inference in Adaptive Network Epidemic Models

Project for the course Simulation (ST3247) taken under A. Prof. Alexander Thiery. · [repo](#)

- Developed simulation-based inference framework for adaptive epidemic network models with *rewiring dynamics*.
- Applied Approximate Bayesian Computation (ABC) for likelihood-free parameter inference under intractable likelihoods.
- Evaluated automatic summary statistic design and calibration quality across inference settings.
- Compared ABC with Neural Posterior Estimation (normalizing flows) for amortized posterior inference

Monte Carlo Methods and their Applications in Finance

Capstone project supervised by Prof. Ren Weiqing. · [repo](#)

- Investigated variance reduction techniques for Monte Carlo estimation in finance.
 - Implemented and compared control variates, importance sampling, and control functionals in rare-event settings.
 - Applied methods to deep out-of-the-money option pricing and credit risk default probability estimation under multi-factor normal copula models.
 - Analyzed estimator efficiency and variance behaviour under extremely low-probability events.
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WORK EXPERIENCE

CS1010S Teaching Assistant

School of Computing · (Aug 2023 - Dec 2023)

- Conducted weekly tutorials for the introductory programming module with positive student feedback (4.2/5.0).
- Guided students during office hours on debugging and computational thinking in Python.

Resident Assistant

Ridge View Residential College · (Aug 2024 - May 2026)

- Maintained a safe and inclusive residential community through structured engagement and student support.
 - Coordinated with residential life staff to respond to crisis situations and ensure timely resolution and follow-up.
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SKILLS

- Programming & Tools: Python, SQL, R, LaTeX, Typst
- MLOps: PySpark, Hadoop, Snowflake
- Scientific Computing: NumPy, SciPy, Pandas, matplotlib, PyTorch and Jax.

Coursera

- SQL for Data Science (UC Davis)
- Spark, Hadoop, and Snowflake for Data Engineering (Duke)
- Bayesian Statistics: From Concept to Data Analysis (UC Santa Cruz)